PROJECT	DOCUMENT	REVISION	DATE
APOLLO TRACK POD	FABRICATION BLUEPRINT	002	2026-07-12

UNITS	SHEETS
MM [IN]	7 + SCHEDULES

MODIFICATION TYPE — ARTICULATED SPLIT-FRAME

ARTICULATED "SPLIT-FRAME" SUSPENSION MOD

Converts a rigid rubber-track pod on a hub-motor scooter (motor inside the Ø180 10T sprocket on a static Ø10 axle between fork legs, two bearing-mounted lower idlers, lugged rubber band) into an independently articulating suspension: a central pivot below the drive hub, a leading and a trailing arm, and two coil-over shocks. Rev 002: every load-bearing dimension is measured and baked in as a number — the only open measurements are the fork-leg thickness and the guide-lug height.

§0 READ THIS BEFORE ANYTHING ELSE

The pod you bought (photos: sprocket + two idlers + rubber band, sold as a wheel-replacement track for mowers / small ATVs / mobility platforms) has **no external steel frame**. The idlers hang off an internal molded/plate structure hidden behind the sprocket, and dimensions vary between sellers even when the listings look identical. That structure is exactly what this mod removes and replaces, so:

- **This drawing set is parametric.** Datums A–J on Sheet 0 are measured off *your* pod with calipers and a ruler. Every dimension on Sheets 1–6 is a formula of those datums. Fill the table in, then transfer numbers to the drawings.
- **Disassemble one pod completely first.** Photograph the internal frame, the idler axle hardware, and — critically — however the pod stops itself from spinning around the drive axle (anti-rotation). Your carrier plates must reproduce that interface.

- **Build one pod, test it, then replicate.** Don't cut four sets of parts before the first pod has survived the Sheet-7 test protocol.

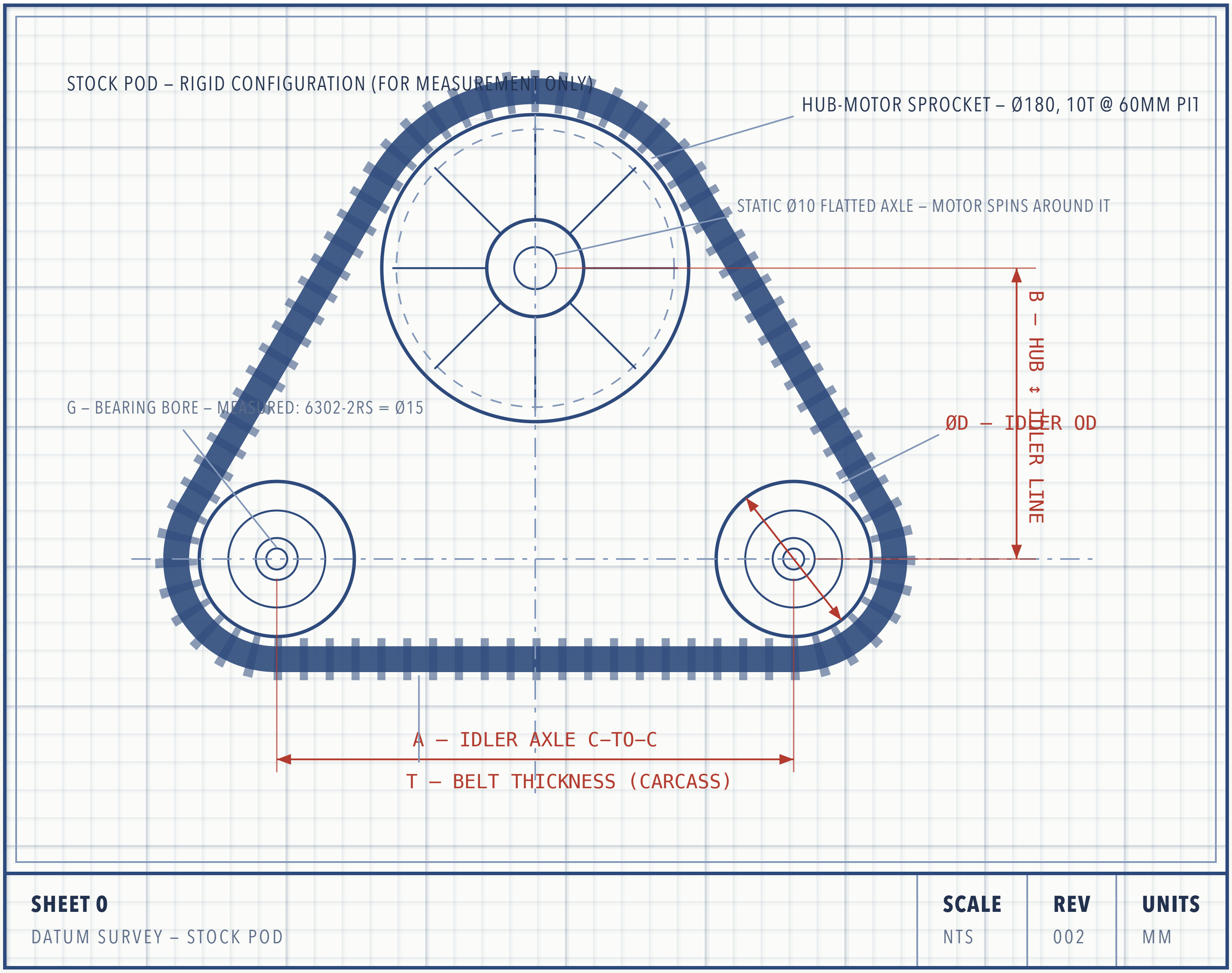
DESIGN RISK – READ

Track retention is the failure mode of this mod. A rigid triangular pod keeps the belt at constant wrap length. Once the arms articulate, the belt path length changes a few millimetres through travel and the belt *will* try to walk off the idlers under side load. The design compensates three ways, and all three are mandatory: travel limited to ±15°, spring preload keeping the belt tensioned through droop, and the trailing-arm tensioner (Sheet 6) set correctly. Deep-lug tracks like yours help; they are not sufficient alone.

Anti-rotation — resolved in Rev 002. The carriers bolt directly to the scooter-fork legs, and their axle slots grip the hub-motor's flatted Ø10 axle exactly like a scooter torque arm. Drive torque reacts straight into the fork; no separate link needed. What replaces it as a watch-item: **the 118 mm belt runs in the front fork's 120 mm gap — 1 mm per side.** Verify the belt tracks true before building anything on the front pod; the rear (140 mm gap) is comfortable.

SHEET 0 DATUM MEASUREMENTS – TAKE THESE OFF YOUR POD

Set the pod on a flat floor, belt tensioned as delivered. Measure with calipers where possible. Record everything in the table; the formula column on later sheets consumes these letters.



DATUM TABLE – FILL IN WITH CALIPERS

DATUM	WHAT TO MEASURE	HOW	TYPICAL*	YOURS
L	Belt cord-line circumference × overall width	Links × pitch off the belt spec	—	**1080 × 118** (60 mm pitch × 18 links, 2.1 kg)
A	Idler axle centre-to-centre distance	Solved from L along the belt *cord line* (10T × 60 = cord Ø191, 5.5	230–300	**222.2** (solved)

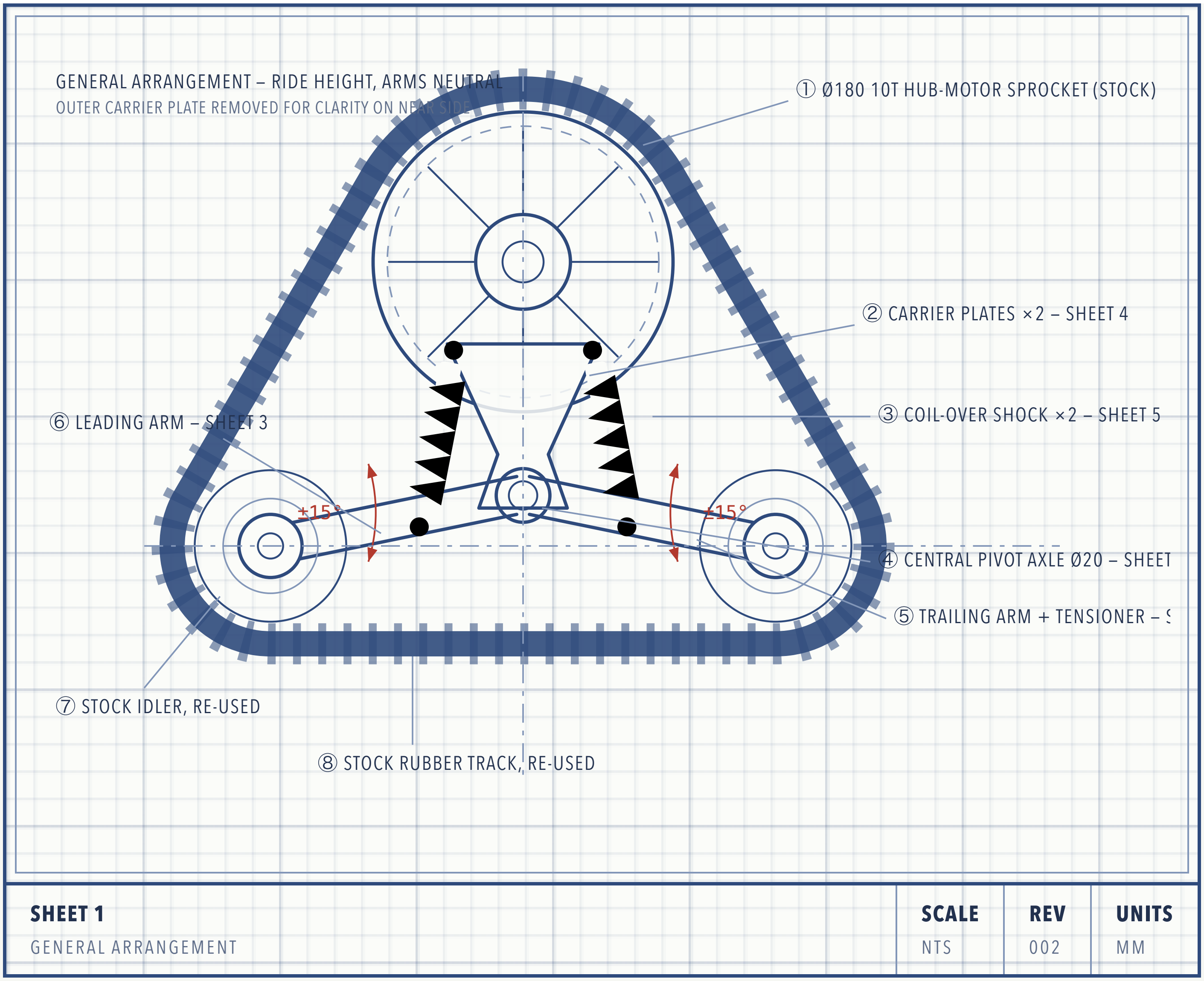
DATUM	WHAT TO MEASURE	HOW	TYPICAL*	YOURS
		above the Ø180 face) — tensioner absorbs the residue		
B	Drive-hub centre height above idler axle line	Square + ruler from floor; B = hub height – idler height	150–200	**170**
D	Idler wheel outside diameter	Calipers across the wheel	90–120	**108** (WJ wheel)
G	Idler bearing bore (axle Ø)	Read bearing number off seal	15–20	**15** — 6302-2RS (15×42×13); caliper read 14.86
fork	Fork-leg inner gap, front / rear pod	Inner face to inner face	—	**120 / 140** — leg thickness ____ (30 assumed)
axle	Hub-motor axle	—	—	**Ø10, flatted, static** — motor spins around it
F	Belt inner width between guide lugs	Derived — wheel and belt are a matched set	60–90	**62** (H + 4); lug height ____ (20 assumed, own

DATUM	WHAT TO MEASURE	HOW	TYPICAL*	YOURS
H	Idler hub width (bearing-to-bearing over the wheel)	Calipers over the wheel hub	40–60	**58** (57.85)
T	Belt carcass thickness (not incl. lugs)	Calipers on a straight span edge	8–14	~**12** (confirmed)

*Typical ranges for this class of pod — for sanity-checking only, never for cutting. Derived values used on later sheets (exact, echoed by the .scad on every render): A = 222.2 · P = B − 38 = 132 · C = $\sqrt{(A/2)^2 + 38^2}$ = 117.4 · neutral arm droop 18.9° · shock station a = 0.68·C = 79.8 · upper shock eye at (±47, −5) from hub centre · keel centre 104 below hub centre · wheel travel +30 / −27.5. The 38 mm pivot lift (was 20, then 30) keeps everything crossing the belt zone clear of the bottom run at full bump; 38 is near the hard limit of 42 where the keel window closes.

SHEET 1 GENERAL ARRANGEMENT – HOW THE MECHANISM WORKS

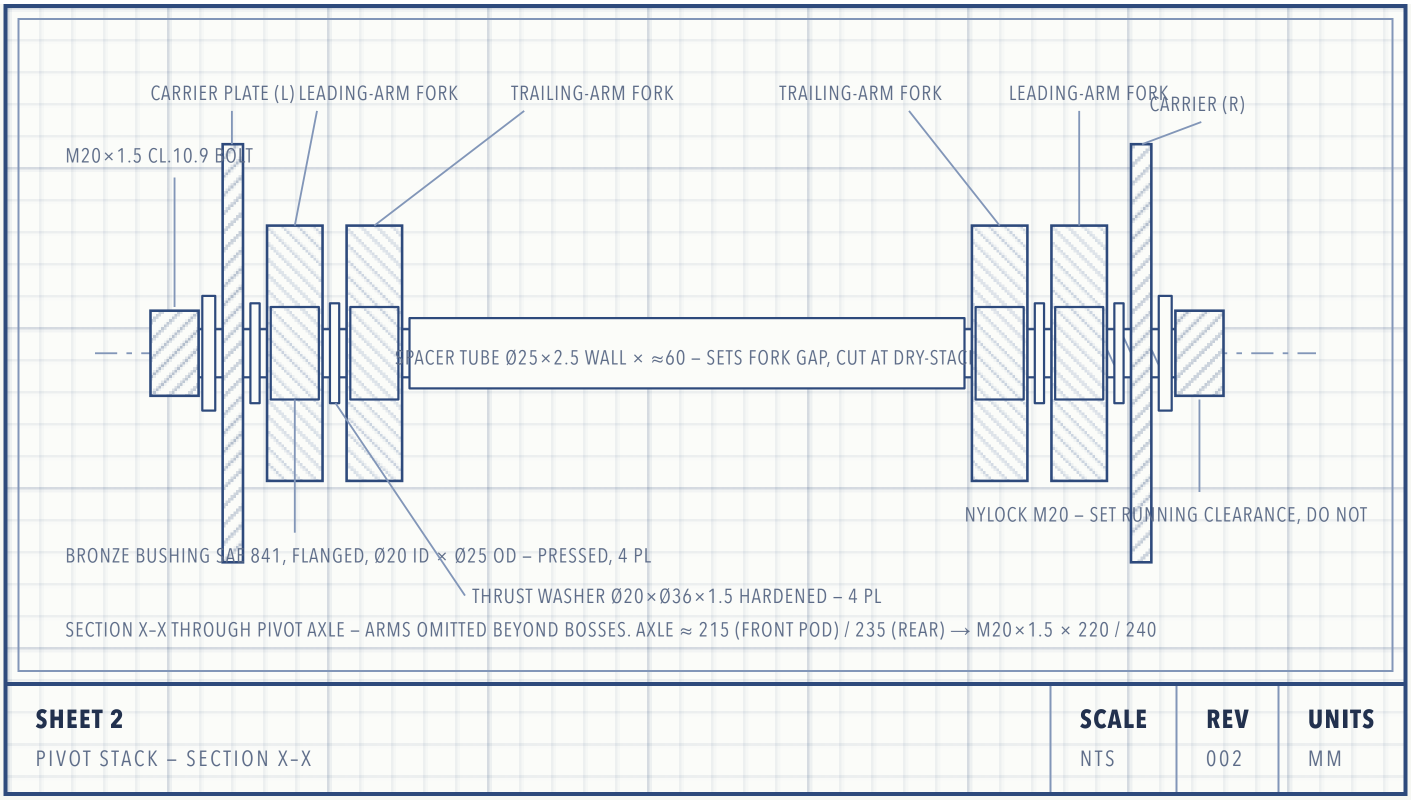
The rigid internal frame is replaced by two identical steel **carrier plates** that hang from the scooter-fork legs: each plate slots over the hub-motor's static Ø10 flatted axle (doubling as a torque arm) and takes one M8 bolt into the leg 52 mm above (Rev 002 — no carrier bearings, no anti-rotation link). A **Ø20 pivot axle** spans the plates 132 mm below the hub. The **leading arm** (front idler) and **trailing arm** (rear idler) are fork-shaped weldments that swing on that axle like scissor halves. Each arm carries a **coil-over shock** reacting against tabs on the carrier inner faces. Wheel hits a bump → arm swings up → shock compresses. Each side is fully independent.



Arms drawn at neutral. Each wheel gets **+30 mm bump / −27.5 mm droop** at ±15° (C = 117.4) — a huge improvement over the rigid pod's zero. At full bump the belt's bottom run rises ~30 mm toward the pivot; everything that crosses the belt zone clears it (pivot spacer 29 mm, keel 62 mm above the assumed lug tops), and the carrier plates themselves ride at |z| = 90 — outside the belt's 59 mm half-width, so they *cannot* touch the track at any articulation. Verify with the OpenSCAD clearance guard (all lines must read PASS).

SHEET 2 CENTRAL PIVOT AXLE – STACK-UP SECTION

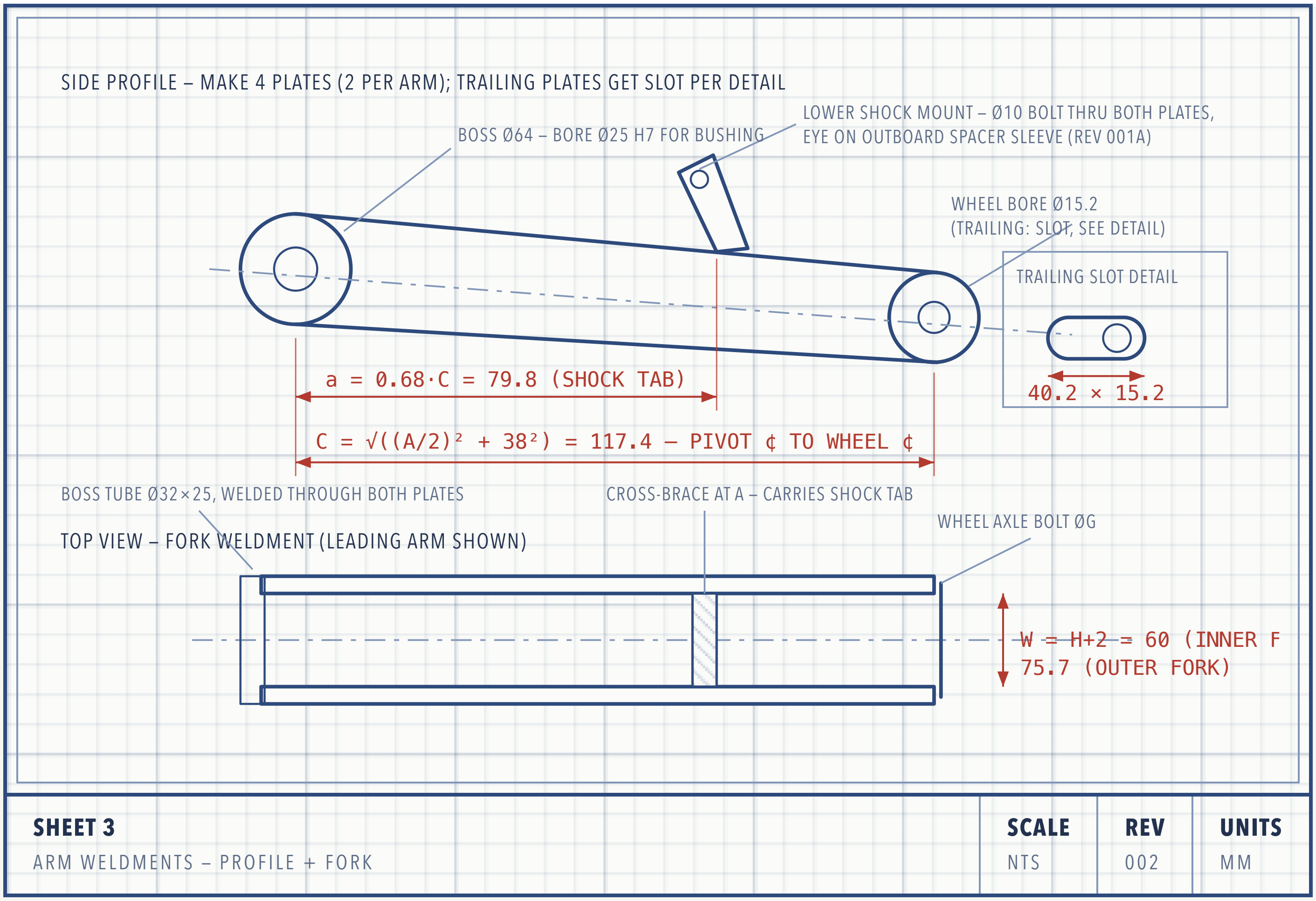
Both arms share one axle. To keep each idler wheel centred on the belt, each arm is a **fork** (two plates): the trailing arm is the narrow inner fork, the leading arm the wider outer fork, nested like scissors. Four bronze bushings — one pressed into each fork plate — carry the rotation. The axle itself is a partially-threaded M20 class-10.9 bolt clamping nothing: the nylock nut is set to leave running clearance, then the spacer tube takes the clamp load.



- Bushing seats in the fork plates: bore Ø25 H7. Press bushings with a vise and a socket, never a hammer. Ream to Ø20 sliding fit after pressing if they close up.
- Grease: lithium EP2 at assembly; add a cross-drilled grease passage + M6 zerk in each fork boss if you want serviceability (recommended — this joint lives in mud).
- The spacer tube length sets the lateral position of both forks: cut it so each idler wheel sits dead-centre in the belt (check against datum F before welding anything).

SHEET 3 ARTICULATING ARMS – LEADING & TRAILING

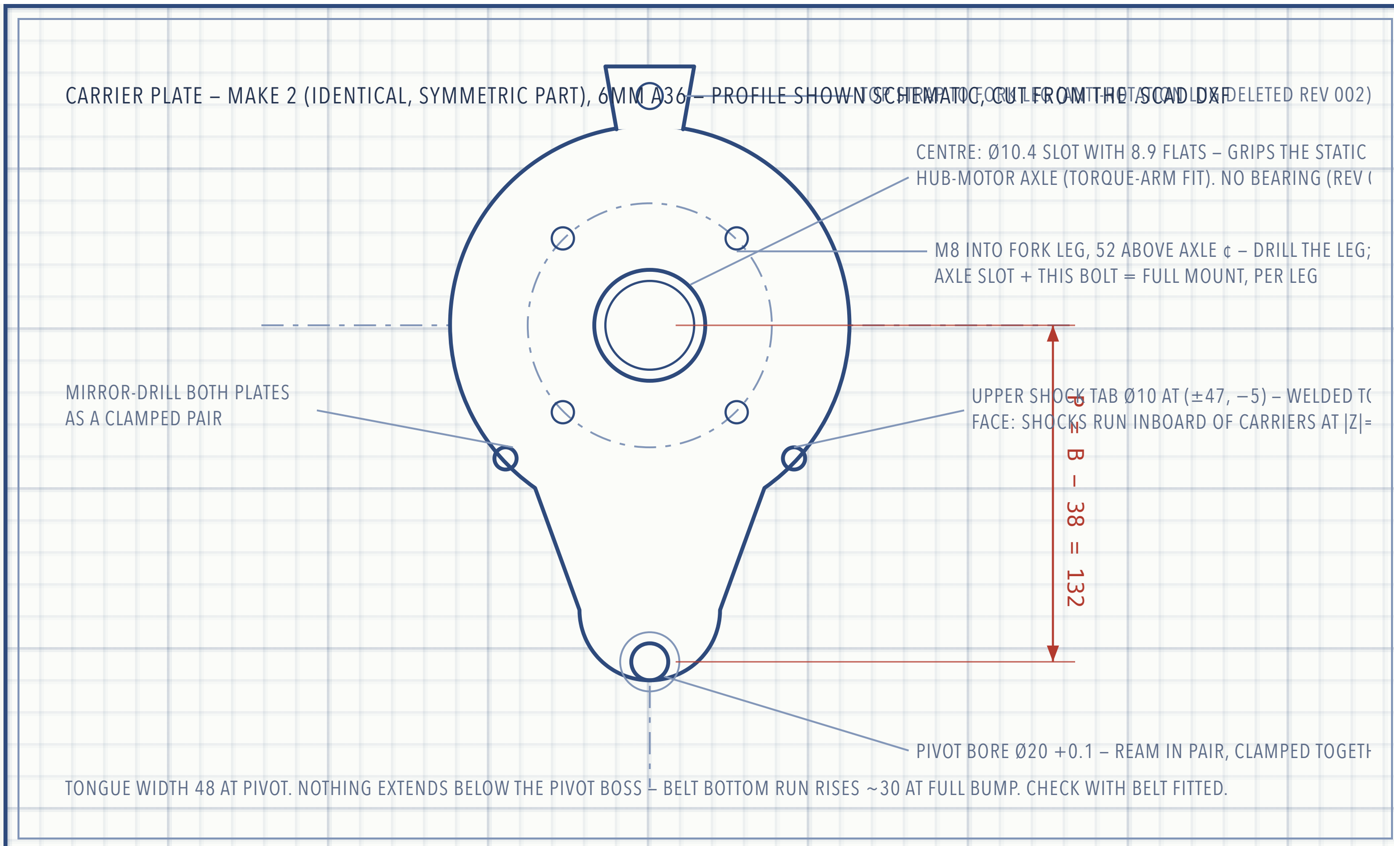
Each arm is a weldment of two profile plates (6.35 mm / ¼″ A36 steel), a bushing boss tube at the pivot end, a cross-brace, and a shock tab on the brace. The two arms are identical *except*: the trailing arm's wheel bore is a **25 mm slot** for the belt tensioner (Sheet 6), and its fork is the narrower, inner one.



- **Arm length C is sacred.** At neutral (arms level) the wheel centres must land exactly where the stock frame held them, or the belt won't tension. Cut plates 1 mm long and file the wheel bores to final position with the belt fitted.
- Plate profile: 64 mm wide at the boss tapering to 52 mm at the wheel end is plenty for a <150 kg vehicle corner. Don't drill lightening holes on rev 001.
- Steel over aluminium: 6061-T6 loses ~40% strength in the weld HAZ and these arms are all welds. Use A36/S235 steel unless you're bolting, not welding.

SHEET 4 CARRIER PLATES – THE NEW BACKBONE

Rev 002: two **identical** 6 mm steel plates (symmetric part — one laser profile serves both sides) hang from the scooter-fork legs, plates on the leg *outer* faces (|z| = 90 front pod / 100 rear). Each has: a **Ø10.4 flatted axle slot** that grips the hub-motor axle like a torque arm, an **M8 hole 52 mm above** it that bolts into the drilled fork leg, the **Ø20 pivot bore 132 mm below** the axle, the **keel M8 at 104 mm below**, and shock-tab weld lobes reaching (±47, −5). The old centre bearing and anti-rotation lug are deleted — the fork does both jobs.



SHEET 4	SCALE	REV	UNITS
CARRIER PLATE – PROFILE	NTS	002	MM

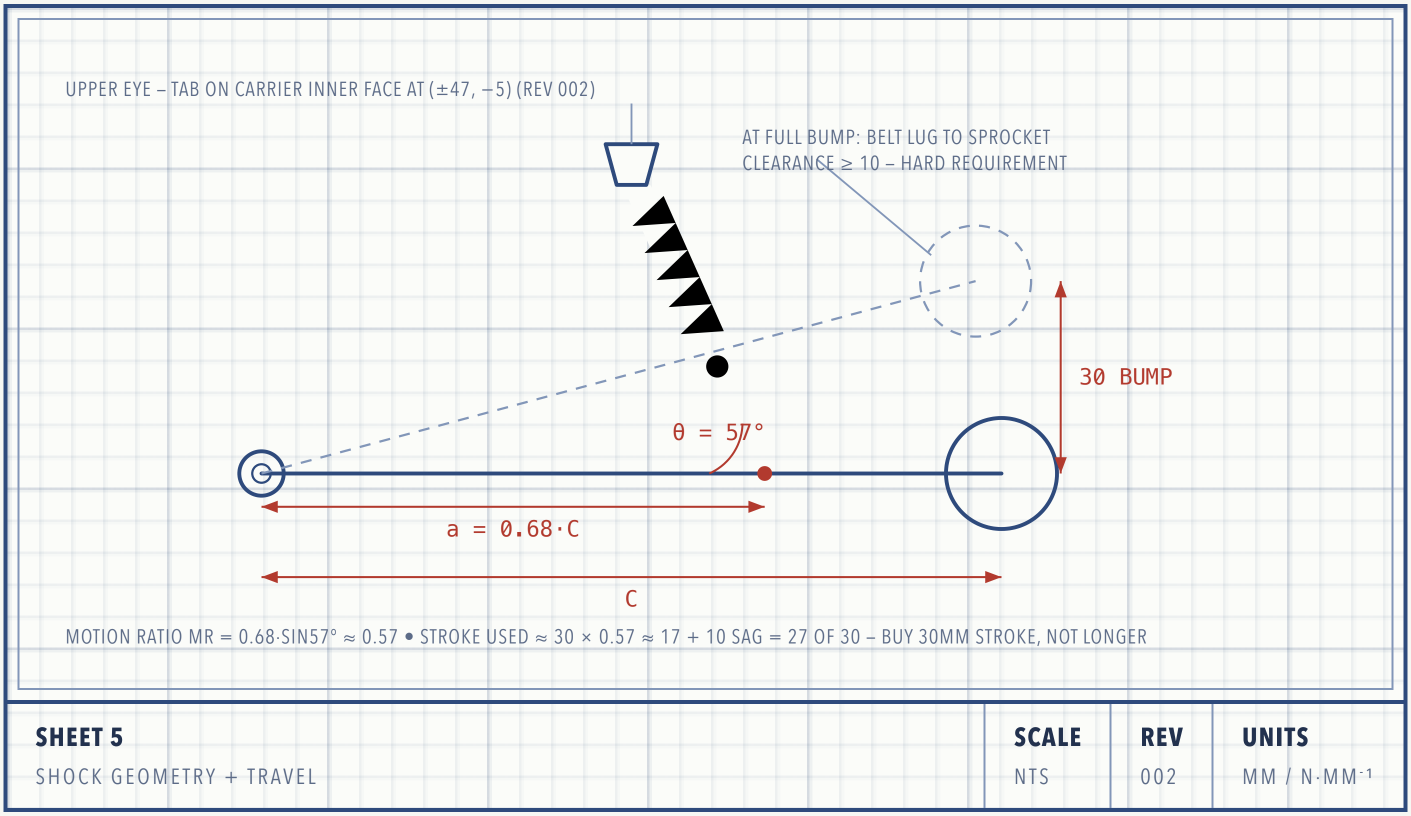
FIT-UP NOTE

The two carrier plates are joined into one rigid box by: the hub-motor axle (top), the pivot axle assembly (middle), and one M8 **keel standoff at 104 mm below the hub centre — between the sprocket's swept disc and the pivot boss** (the 001a keel below the pivot would have hit the rising belt at full bump by ~40 mm; this window clears the sprocket by 6 mm and the pivot washers by 2 mm). Bolt both plates to the fork legs before drilling the keel holes so the box squares itself to the fork. The motor must spin freely with ≥3 mm side clearance each face.

The companion OpenSCAD model prints a **PASS/WARN belt-clearance line for every hanging part** (tongue, keel, pivot spacer, boss) at full bump — all must PASS with ≥5 mm before cutting steel, and the check must be re-run once the belt's guide-lug height is measured (datum F row).

SHEET 5 SHOCK GEOMETRY, TRAVEL & SPRING SIZING

The shock mounts at a $= 0.68 \cdot C$ along the arm, inclined 55–60° to the arm. That placement is deliberate — it sets the motion ratio high enough that off-the-shelf 150 mm e-scooter/pit-bike coil-overs work without absurd spring rates.



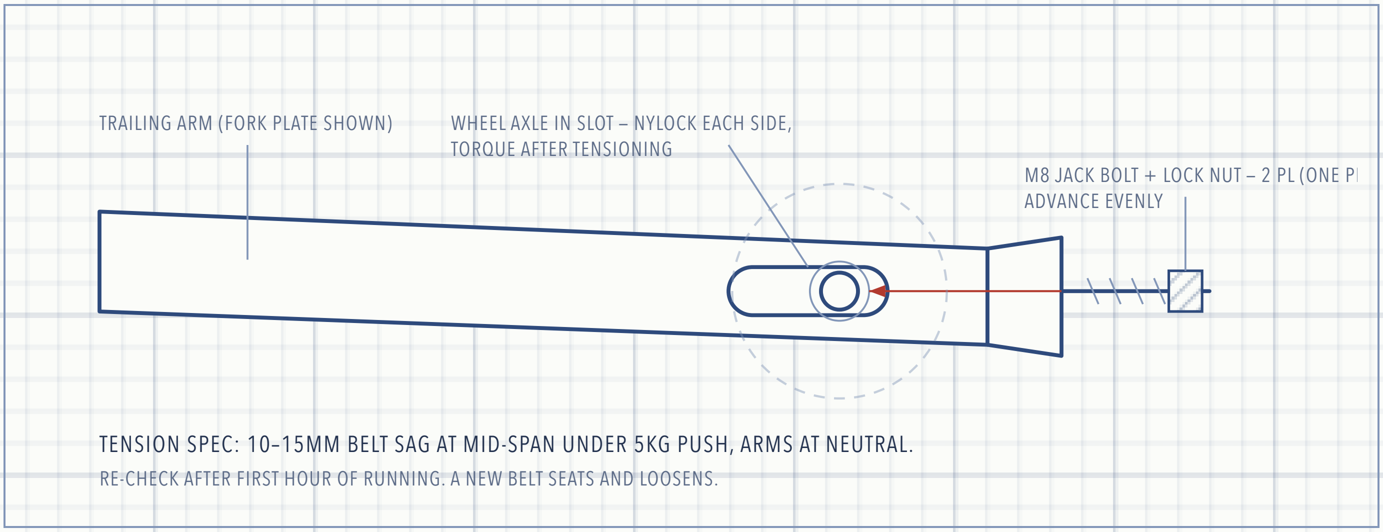
SPRING RATE – WORKED METHOD

1. Weigh the corner: put the vehicle's wheel-position on a bathroom scale, loaded (rider + cargo). Call it `M` kg → force `F = 9.81·M` N. That corner load splits across the pod's two arms roughly 50/50 → `F_arm = F/2`.
2. Wheel rate target: static sag should be 25–30% of bump travel. Measured pod (C = 117.4) → bump travel 30 mm, sag ≈ 9 mm → `k_wheel = F_arm / 9` N/mm.
3. Spring rate: `k_spring = k_wheel / MR²` with MR ≈ 0.58 → `k_spring ≈ 3.0 × k_wheel`.

Example — 160 kg loaded vehicle on 4 pods → M = 40 kg, F = 392 N, F_arm = 196 N → k_wheel ≈ 22 N/mm → **k_spring ≈ 67 N/mm ≈ 385 lb/in**. That's a heavy e-scooter / mini-moto spring. Buy adjustable-preload coil-overs, **150 mm eye-to-eye, 30 mm stroke (not longer — Sheet 5 stroke budget)**, and buy one spare spring one step stiffer and one softer. Weigh the real corner before ordering springs.

SHEET 6 BELT TENSIONER – TRAILING ARM

The articulating geometry cannot hold belt tension by luck; the trailing wheel axle rides in the Sheet-3 slot and is pushed rearward by a jack bolt against a welded lug. This gives ~25 mm of tension range — enough to fit the belt slack and take up wear.



SHEET 6	SCALE	REV	UNITS
TENSIONER – TRAILING AXLE	NTS	002	MM

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§ 7 BILL OF MATERIALS

#	ITEM	QTY/POD	SPECIFICATION
1	Pivot axle	1	M20×1.5 × 220 (front pod) / 240 (rear) cl.10.9, part-threaded
2	Arm profile plates	4	6.35mm (¼″) A36/S235 steel
3	Carrier plates	2	6mm A36/S235, mirrored pair

#	ITEM	QTY/POD	SPECIFICATION
4	Boss tubes	2	Ø32 OD × Ø25 ID DOM tube, 25 long
5	Bronze bushings	4	SAE 841 flanged, Ø20 ID × Ø25 OD × 20
6	Thrust washers	4	Ø20 × Ø36 × 1.5, hardened
7	Spacer tube	1	Ø25 OD × 2.5 wall × ≈60
8	Coil-over shocks	2	150mm eye-to-eye, 30mm stroke, adj. preload
9	Shock mounts	2+2	2 upper tabs 6mm w/ Ø10 hole; 2 lower spacer sleeves Ø15×Ø10 ×
10	Fork mount hardware	2+2	Hub-motor axle nuts M10 (stock) + M8×45 cl.10.9 + nylock per l

#	ITEM	QTY/POD	SPECIFICATION
11	Keel standoff	1	Ø16 tube × 180 + M8 × 205 threaded rod cl.10.9
12	Jack bolts	2	M8×40 + lock nuts
13	Anti-rotation link	—	DELETED rev 002
14	Hardware kit	—	Cl.10.9 (Gr8) bolts, nylock nuts, washers
15	Grease fittings	2	M6 zerk + EP2 lithium grease
16	Idler axles	2	Ø15 — 15×100 MTB thru-axle (M15×1.5) or 15 mm shaft + collars
—	Re-used from stock pod	—	Hub-motor sprocket (Ø180, 10T), 2 idlers (Ø108×58), belt (1080

§ 8 FABRICATION SPECIFICATIONS

CUTTING	Laser/waterjet preferred (send Sheet 3/4 profiles as DXF once dimensioned). Hand fabrication acceptable: plasma or 1 mm cutoff wheel, finish edges to bright metal, drill press only for bores — no hand drilling of bearing/bushing seats.
CRITICAL BORES	Bushing seats Ø25 H7 (step-drill then ream). Pivot bores in carrier plates reamed as a clamped pair so the axle is square. Wheel bores ØG +0.2.
WELDING	MIG, ER70S-6, 5 mm fillets. Boss tubes welded both faces. Shock tabs welded both sides — a single-side tab weld is the classic failure. No preheat needed on A36 at these thicknesses. Weld with bushings *out* (heat kills them); press bushings after paint.
SQUARENESS	Fork plates must be parallel within 0.5 mm over their length, or the idler runs angled and throws the belt. Tack, measure, weld in a staggered sequence, re-measure.
FINISH	Degrease, zinc-rich primer, topcoat. Mask every bore and bearing seat.
TORQUE	M8 cl.10.9 — 30 N·m · M10 — 60 N·m · M12 — 105 N·m · M20 pivot nylock — snug to zero end-float then back ⅛ turn (running fit, not clamped). Mark every fastener with paint-pen witness lines.

§ 9 ASSEMBLY SEQUENCE

1. **STRIP THE DONOR POD.** Remove belt, idlers, sprocket. Photograph and measure the internal frame and anti-rotation feature (datum J) before discarding it.
2. **PRESS BUSHINGS.** Four flanged bronze bushings into the fork-plate bosses (vise + socket). Verify the Ø20 axle turns by hand with no rock.
3. **DRY-STACK THE PIVOT.** Thread onto the axle: carrier L → washer → leading fork → washer → trailing fork → spacer tube → trailing fork → washer → leading fork → washer → carrier R (Sheet 2 order). Cut the spacer so each idler centreline lands mid-belt (datum F). No welding yet.
4. **HANG AND BOX THE CARRIERS.** Slot both plates onto the hub-motor axle, bolt the M8 into each drilled fork leg, then fit the keel standoff (drill its holes with the plates bolted up so the

box squares to the fork). Spin the motor — ≥3 mm clearance each face, zero rub, and check the belt's 1 mm/side gap to the front fork legs.

5. **LOCATE SHOCK MOUNTS.** With arms level and shocks held in place at ~30% compression, mark positions: upper tabs on the carrier *inner* faces at (±47, −5), lower eyes on Ø10 through-bolts with spacer sleeves at a = 79.8, θ = 57° (shocks run inboard of the carriers at |z| = 73 — rev 002). Tack the tabs, articulate ±15° by hand checking nothing binds, then final-weld with bushings removed.
6. **PAINT, THEN FINAL-ASSEMBLE** the pivot stack with grease, set the M20 nylock for free running, fit shocks.
7. **FIT IDLERS** to arm forks — leading in the round bore, trailing in the slot, jack bolts backed off fully.
8. **WRAP THE BELT.** Sprocket first, engage lugs, lever over the idlers with the trailing axle at its most-forward slot position (rubber levers only — no screwdrivers into the carcass).
9. **TENSION.** Advance both jack bolts evenly to 10–15 mm mid-span sag, square the axle in the slot (equal bolt counts), torque the axle nylocks.
10. **FINAL TORQUE.** Motor axle nuts + M8 fork bolts + pivot nylock witness-marked, then run the §10 test protocol before carrying a rider.

§10 TEST PROTOCOL & SAFETY

1. **Bench articulation:** pod off the ground, cycle each arm full travel 50× by hand. No binding, no belt-lug contact with the sprocket at full bump (≥10 mm, Sheet 5), no shock coil-bind.
2. **Bench spin:** drive the pod unloaded at walking-pace RPM for 5 minutes. Watch belt tracking — it must not walk toward either edge. If it walks, the fork plates aren't parallel or the tension axle isn't square.
3. **Loaded creep:** vehicle weight on, no rider, walk-beside at <5 km/h over a 50 mm plank. Confirm articulation and belt retention.
4. **Ridden test:** flat ground, low speed, then progressively rougher terrain. Stop and inspect after 10 minutes: retorque everything, re-check belt tension (new belts loosen), look for cracked tack welds and shiny rub marks.
5. **Retorque schedule:** after 1 h, 5 h, then every 20 h. Witness-mark lines make this a 2-minute glance.

SAFETY

Tracked pods are finger-traps: never work near a powered, moving belt; the sprocket-to-belt nip point will pull a glove in instantly. Wear eye protection when levering the belt on — it stores real energy. And treat rev 001 as a prototype: first rides in an open area, helmet on, away from traffic and bystanders.

§11 GENERAL NOTES

- All dimensions mm unless noted. Formulas reference Sheet-0 datums; verify every derived number against the physical pod before cutting.
- Break all sharp edges 0.5 mm. Deburr every bore both sides.
- Fasteners class 10.9 (metric) / Grade 8 (US) minimum at pivot and shock points; all nuts nylock or thread-locked.
- This drawing set describes one pod. The carrier plate is a symmetric part — the same profile serves both sides and both pods; only the pivot/keel lengths differ between the front pod (fork gap 120) and rear pod (fork gap 140).
- Rev 001 supersedes the AI-concept sketch it was derived from. Known deltas: arms changed from single plates to fork weldments (keeps idlers centred on the belt), shock tab moved from 0.45·C to 0.68·C (sane spring rates), tensioner changed from eccentric cam to slot + jack bolt (easier to fabricate accurately), carrier bearing + anti-rotation link added (the concept had no torque path to the chassis).
- **Rev 001a** (from the 3D fit-check in the companion OpenSCAD model): shocks moved outboard of the carrier plates and the four cross-standoffs replaced by a single keel standoff below the pivot — both original placements collided with the sprocket's swept volume.
- **Rev 001b** (measured datums, 2026-07-12): track 1083×118 (60 mm pitch × 18 links) — idler spacing A now solved from belt length (≈ 243); sprocket OD 180; idlers Ø108×58 on 6302-2RS bearings → Ø15 axles. **Belt-clearance fix:** at full bump the belt bottom run rises ~32 mm and the 001a keel would have collided by ~40 mm — keel relocated between sprocket and pivot boss, pivot raised (P = B – 30), carrier tongue slimmed Ø56→Ø48. The OpenSCAD model prints a PASS/WARN clearance check per hanging part on every render.
- **Rev 001c**: pivot raised further (P = B – 38, near the 42 mm limit where the keel window closes) — lug-top margin at full bump grows 7.6→15.3 mm and the steel-touches-belt angle moves from ~+18.5° to ~+22°, buffering against shock-travel overshoot. Keel moved up 6 mm

with it; keel-to-sprocket and keel-to-pivot-washer gaps added to the clearance guard. Cost accepted: belt-path variation through travel grows 3.6→5.7 mm, within tensioner range.

- **Rev 002** (owner interview): the pod is a **hub-motor scooter conversion** — motor inside the Ø180 10T sprocket on a static Ø10 flatted axle between fork legs (gaps: front 120, rear 140). Carriers redesigned as fork-hung torque-arm plates (identical symmetric part; axle slot + one M8 per leg); 6205 bearings, axle sleeve and anti-rotation link deleted; shocks moved inboard of the carriers ($|z| = 73$). Belt solver corrected to the cord line (10T × 60 = Ø191): A = 243 → 222.2, C = 117.4. Carriers now ride outside the belt width — the tongue-hits-track failure mode is geometrically impossible. New fit risk: 118 belt in the 120 front fork gap (1 mm/side).
